

CHAPTER 11



Class : 12th Biology

Topic : Ecology

For : NEET UG

Organisms and Populations

* WHAT IS ECOLOGY?

Ecology is the study of interactions among organisms and their **Environment**.

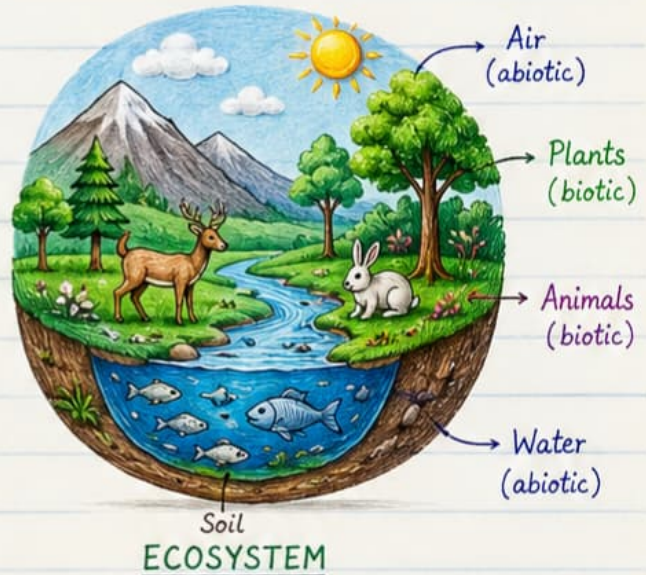
It includes both living (**biotic**) and non-living (**abiotic**) factors and how they influence each other.

Ecology is the biology of home.

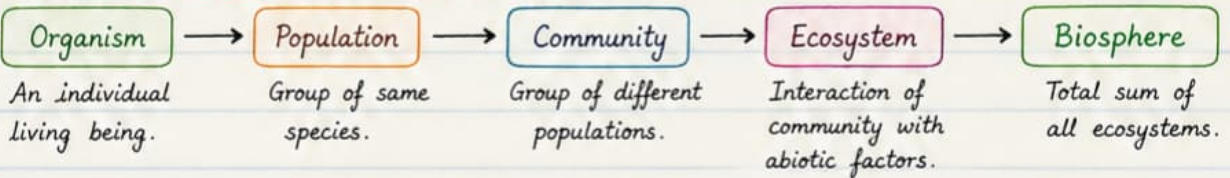
— Earnst Haeckel

* KEY CONCEPTS IN ECOLOGY

- **Organism** — Individual living being.
- **Population** — Group of similar organisms in a particular area.
- **Community** — Different populations living and interacting in an area.
- **Ecosystem** — Community + abiotic environment interacting as a functional unit.
- **Biosphere** — The sum total of all ecosystems on Earth.



* LEVELS OF ORGANISATION IN NATURE



* BIOTIC AND ABIOTIC FACTORS

BIOTIC FACTORS (Living Components)

- Include all living organisms.
- Examples: Plants, animals, fungi, bacteria, humans.



ABIOTIC FACTORS (Non-living Components)

- Include physical and chemical factors.
- Examples: Light, temperature, water, soil, air, minerals, pH, humidity.



* POPULATION

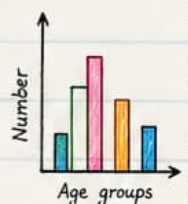
A population is a group of individuals of the **same species** living in a specific area at a specific time and capable of **interbreeding**.

Example: All the deer in a forest.



POPULATION CHARACTERISTICS

- ★ Birth rate (natality)
- ★ Death rate (mortality)
- ★ Sex ratio
- ★ Age distribution
- ★ Population density



Study Nature, Love Nature, Stay Close to Nature. ♥

POPULATION – KEY POINTS

- Members of a population share a **common gene pool**.
- Population shows certain characteristics like **birth rate, death rate, density** etc.
- It is the **basic unit** in ecology.
- Populations are **dynamic** and change with time.
- Example** : Paramecium in a pond, Humans in a city.

POPULATION DENSITY

Number of individuals of a species per unit **area** or **volume**.

FORMULA

$$\text{Population density} = \frac{\text{Number of individuals}}{\text{Unit area or volume}}$$

TYPES OF POPULATION DISTRIBUTION

CLUMPED

Individuals occur in groups or clusters.



UNIFORM

Individuals are evenly spaced.



RANDOM

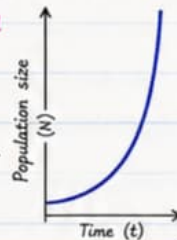
Individuals are irregularly distributed.



GROWTH PATTERNS IN POPULATIONS

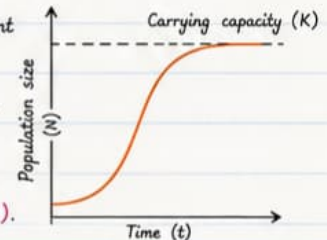
1. EXPONENTIAL GROWTH (J-SHAPED CURVE)

- Occurs in an environment with **unlimited** resources.
- Population size increases rapidly.
- Example** : Bacteria in a nutrient medium.



2. LOGISTIC GROWTH (S-SHAPED CURVE)

- Occurs in a natural environment with **limited** resources.
- Population growth slows down and stabilizes.
- Population reaches a stable level called **carrying capacity (K)**.



FACTORS AFFECTING POPULATION GROWTH

1. NATALITY (BIRTH RATE)

Increase in the number of individuals by birth.



2. MORTALITY (DEATH RATE)

Decrease in the number of individuals by death.



3. IMMIGRATION

Movement of individuals into a population.



4. EMIGRATION

Movement of individuals out of a population.



INTERACTIONS AMONG ORGANISMS

TYPE	DESCRIPTION	EXAMPLE	DIAGRAM
1. COMPETITION	Two organisms compete for the same resources.	Plants competing for sunlight.	
2. PREDATION	One organism (predator) kills and eats another (prey).	Lion and Deer	
3. PARASITISM	Parasite lives on or in the host and harms it.	Tapeworm in human intestine	
4. MUTUALISM	Both the organisms benefit.	Lichen (Algae + Fungus)	
5. COMMENSALISM	One benefits, the other is neither helped nor harmed.	Orchid on a tree	

ECOLOGICAL PYRAMIDS

1. PYRAMID OF NUMBERS



2. PYRAMID OF BIOMASS



REMEMBER



Population is the basic unit in ecology, Its size is controlled by birth rate, death rate, immigration and emigration.

Small steps in understanding nature lead to big success in **NEET**!



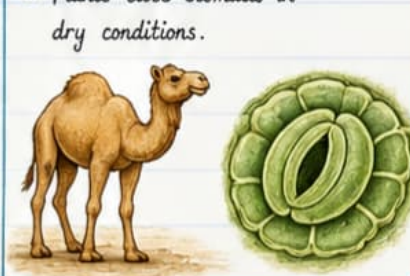
ADAPTATIONS IN ORGANISMS

Adaptation is any **structural**, **behavioural** or **physiological** feature that helps an organism to survive in its habitat.

- Example : Thick fur in polar bear, spines in cactus, broad leaves in aquatic plants.

TYPES OF ADAPTATIONS

- Structural (Morphological)** - Physical features.
- Behavioural** - Actions or behaviour.
- Physiological** - Internal body functions.

1. STRUCTURAL (Morphological)	2. BEHAVIOURAL	3. PHYSIOLOGICAL
Changes in body parts or shape. Examples : - Cactus has spines to reduce water loss. - Duck has webbed feet for swimming.	Changes in actions or behaviour of an organism. Examples : - Birds migrate to distant places in winter. - Nocturnal animals are active at night.	Changes in body functions or metabolism. Examples : - Camel conserves water by producing concentrated urine. - Plants close stomata in dry conditions.
		

POPULATION INTERACTIONS

Different populations interact with each other in many ways. These interactions can be **positive (+)**, **negative (-)** or **neutral (0)**.

1. COMPETITION (-/-)

Both species are harmed as they compete for the same resource.

Example : Weeds and crop plants compete for nutrients, water and light.



2. PREDATION (+/-)

One organism (predator) benefits and the other (prey) is harmed.

Example : Lion (predator) hunts deer (prey).



3. PARASITISM (+/-)

Parasite benefits and host is harmed.

Example : Cuscuta (dodder) absorbs nutrients from host plant.



4. MUTUALISM (+/+)

Both species benefit from the interaction.

Example : Lichen (algae + fungi) benefit together.



5. COMMENSALISM (+/0)

One species benefits and the other is neither harmed nor helped.

Example : Remora fish gets food from shark while shark is unaffected.



6. AMENSALISM (-/0)

One species is harmed while the other is unaffected.

Example : Penicillium fungus inhibits bacterial growth.



IMPORTANCE OF STUDYING POPULATIONS

- Helps in understanding the size, density and distribution of populations.
- Useful in conservation of endangered species.
- Helps to maintain ecological balance.
- Important for planning and management of natural resources.

QUICK RECAP

- Population = group of same species in a specific area.
- Population density = No. of individuals per unit area or volume.
- Interactions shape the ecosystem.
- Adaptations help organisms to survive.



CHAPTER 11 : ORGANISMS AND POPULATIONS

For NEET UG

POPULATION REGULATION

Populations are regulated by a combination of **density-dependent** and **density-independent** factors. These factors keep the population size in **balance** with the environment.

FACTORS AFFECTING POPULATION SIZE

DENSITY-DEPENDENT FACTORS

Effect increases with increase in population density.

DENSITY-INDEPENDENT FACTORS

Effect remains the same irrespective of population density.

DENSITY-DEPENDENT FACTORS

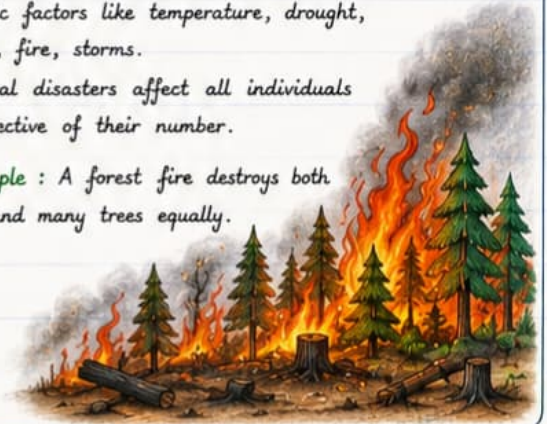
- Competition for food, water, space and light.
- Predation - more prey attracts more predators.
- Parasitism and disease - spread faster in crowded population.
- Territoriality.
- Accumulation of waste products.

Example : In a crowded pond, fish compete for food; some become weak and die.



DENSITY-INDEPENDENT FACTORS

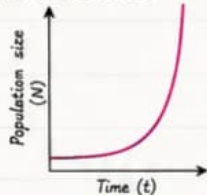
- Abiotic factors like temperature, drought, flood, fire, storms.
- Natural disasters affect all individuals irrespective of their number.
- **Example :** A forest fire destroys both few and many trees equally.



POPULATION GROWTH MODELS

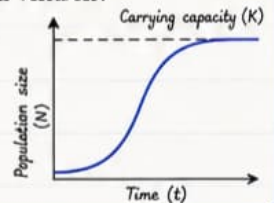
1. J-SHAPED (EXPONENTIAL GROWTH)

- Occurs in an environment with unlimited resources.
- Population grows very fast.
- **Example :** Bacteria in a culture medium.



2. S-SHAPED (LOGISTIC GROWTH)

- Occurs in an environment with limited resources.
- Growth is slow at first, then rapid, then slows down and stabilizes at carrying capacity (K).
- **Example :** Human population, deer in a forest.



METHODS OF STUDYING POPULATIONS

METHOD	DESCRIPTION
1. Quadrat method	Area is divided into quadrats (squares). Organisms in each quadrat are counted and average is calculated. 
2. Transect method	A line (transect) is laid across the area. Organisms on or near the line are counted. 
3. Mark-recapture method	Some individuals are captured, marked and released. After some time, they are captured again. Proportion of marked individuals helps estimate population size. 

IMPORTANCE / APPLICATIONS

- Helps in conservation and management of wildlife.
- Useful in pest control in agriculture.
- Helps in planning sustainable utilization of resources.
- Provides data for environmental impact assessment.
- Essential for maintaining ecological balance.

When we protect populations, we protect the future. ❤️

THINK & REMEMBER

- ✓ Population = group of same species in a specific area.
- ✓ Population density = Number of individuals per unit area.
- ✓ Growth is controlled by both biotic and abiotic factors.
- ✓ Balance in nature depends on stable populations.

+ SUMMARY

- Organisms live in groups called populations.
- Population size changes due to birth, death, immigration and emigration.
- Population growth can be exponential (J-shaped) or logistic (S-shaped).
- Biotic interactions and abiotic factors regulate population size.
- Studying populations helps in conservation and sustainable management of resources.



Study Nature, Understand Life, Protect Earth. ❤️